COURSEWORK ASSIGNMENT SUBMISSION

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Subject: Software Engineering & Project Management **Day:** Day 13 — Software Requirements Specification (SRS)

Section 13: Classroom Activity — Create a Mini SRS

Structured mini Software Requirements Specification (SRS) document for the College Event Management System (CEMS).

1. Purpose & Scope

This document specifies the software requirements for Release 1.0 of the College Event Management System (CEMS). The system manages campus event listings, role-based approval workflows, student registrations, automated ticketing, and real-time gate attendance verification. Out-of-Scope: Third-party commercial payment gateway processing and automated bus transportation scheduling.

2. User Classes & Characteristics

- **Student Users:** General campus student body; access via smartphones; requires intuitive, low-bandwidth UI with minimal data entry.
- **Event Coordinators:** Student club heads and technical leads; create event drafts, monitor seat quotas, and export attendance sheets.
- **Faculty Approvers:** Academic staff and Department Heads; review event safety, resource allocation, and grant formal publication approvals.

3. Core Requirements & Verification Criteria

Req ID	Requirement Description	Priority	Verification Method
CEMS-FR-01	System shall display all published campus events categorized by technical and cultural tags.	High	Inspection & System UI Test
CEMS-FR-02	System shall authenticate users using institutional Single Sign-On (Google OAuth 2.0).	High	Security Integration Test
CEMS-FR-03	System shall atomically reserve a seat and decrement availability upon valid student submission.	High	Automated Concurrency Test
CEMS-FR-04	System shall generate an encrypted, single-use QR ticket upon successful	Medium	QR Decoder Integrity Test

	registration.		
CEMS-FR-05	System shall allow coordinators to record attendance by scanning attendee QR tickets.	Medium	Functional Gate Scan Test
CEMS-NFR-01	System shall process registration transactions in under 1.5 seconds under 200 concurrent users.	High	Performance Stress Benchmark
CEMS-NFR-02	System shall maintain 99.5% service availability across the academic semester.	High	Cloud Monitoring SLA Audit

Section 16: Homework — Formal SRS Document Expansion

Comprehensive architectural expansion of the CEMS Software Requirements Specification.

1. General Constraints & Assumptions

- **Technical Constraint:** The portal backend must run in Linux Docker containers and integrate with existing campus LDAP/SSO directories without schema modification.
- **Policy Constraint:** Student contact data must comply with campus privacy charters; personal phone numbers must be masked from student volunteers.
- **Operational Assumption:** Campus event venues possess minimum 4G mobile coverage or institutional Wi-Fi access at entry checkpoints.

2. Requirements Traceability Matrix (RTM)

Req ID	Business Objective	Design Module	Source Code Component	Test Case ID
CEMS-FR-01	Campus Event Visibility	Catalog UI Module	`EventListController.js`	TC-CAT-01
CEMS-FR-02	Secure Authentication	Auth Service	`OAuthSecurityConfig.java`	TC-SEC-04
CEMS-FR-03	Eliminate Overbooking	Registration Engine	`BookingService.java`	TC-CONCUR-02
CEMS-FR-04	Fast Gate Entry Check	Ticketing Module	`QRCodeGenerator.py`	TC-GATE-01
CEMS-FR-05	Accreditation Records	Reporting Service	`ExportReportService.ts`	TC-REP-03

Peer Review Notes: Reviewed by Peer Team B. Revisions incorporated: (a) Added specific cancellation cutoff window (24h); (b) Specified QR code encryption payload format to prevent barcode forgery.